

## Milestone 3

### Objective: Movie Revenue Prediction

The goal of this project is to develop an **Interactive dashboard** based on machine learning model that accurately predicts a movie's box office revenue based on various features such as budget, genre, director, cast, critic/audience ratings, and historical box office performance.

Our aim is to identify major factors that influence a film's success at the Box Office, by leveraging data from sources like IMDb, TMDb, and Rotten Tomatoes.

### Tech Stack:

- **Python:** Main programming language for data processing & modeling
- **Pandas, NumPy, SciPy, sklearn:** Data manipulation and preprocessing
- **SQLite:** Store cleaned and structured data
- **Matplotlib, Seaborn, Plotly:** Data visualization and EDA
- **Scikit-Learn, Pytorch:** Machine Learning
- **Flask:** API for serving predictions
- **Streamlit:** Interactive web-based dashboard for model insights

**Audience:** filmmakers, producers, and investors.

### Project Demo:

[Movie Revenue Prediction Project Demo Link \(click here\)](#)

### Tool Demo:

[Movie Revenue Prediction Tool Demo Link \(click here\)](#)

### Dataset Description:

| Explored By   | Dataset                | Source | Variables                                                                                                                                                                                                                                                                                      | Dimension                |
|---------------|------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Mohammad Wael | TMDb                   | Kaggle | id, title, vote_average, vote_count, status, release_date, revenue, runtime, adult, backdrop_path, budget, homepage, imdb_id, original_language, original_title, overview, popularity, poster_path, tagline, genres, production_companies, production_countries, spoken_languages, keywords    | 1181912 Rows, 24 Columns |
|               | Rotten Tomatoes Critic | Kaggle | rotten_tomatoes_link, critic_name, top_critic, publisher_name, review_type, review_score, review_date, review_content                                                                                                                                                                          | 1130017 Rows, 8 Columns  |
|               | Rotten Tomatoes Movie  |        | rotten_tomatoes_link, movie_title, movie_info, critics_consensus, content_rating, genres, directors, authors, actors, original_release_date, streaming_release_date, runtime, production_company, tomatometer_status, tomatometer_rating, tomatometer_count, audience_status, audience_rating, | 17712 Rows, 22 Columns   |

|                 |                    |        |                                                                                                                  |                          |
|-----------------|--------------------|--------|------------------------------------------------------------------------------------------------------------------|--------------------------|
|                 |                    |        | audience_count, tomatometer_top_critics_count, tomatometer_fresh_critics_count, tomatometer_rotten_critics_count |                          |
|                 | Title.Basics       | IMDb   | tconst, titleType, primaryTitle, originalTitle, isAdult, startYear, endYear, runtimeMinutes, genres,             | 11464895 Rows, 9 Columns |
| Vishal K. Setti | Name.Basics        |        | nconst, primaryName, birthYear, deathYear, primaryProfession, knownForTitles                                     | 14195120 Rows, 6 Columns |
|                 | Title.Rating       |        | tconst, averageRating, numVotes                                                                                  | 1535911 Rows, 3 Columns  |
|                 | Title.Principals   |        | tconst, ordering, nconst, category, job, characters                                                              | 90984102 Rows, 6 Columns |
|                 | Title.Crew         |        | tconst, directors, writers                                                                                       | 11464895 rows, 3 Columns |
|                 | ISO2 codes to ISO3 | Kaggle | country, alpha2, alpha3, numeric                                                                                 | 249 Rows, 4 Columns      |

Table 1 Dataset Description

### Evaluation and Interpretation

#### 1. Model Performance on Test Set:

We evaluated our Linear Regression, Linear Regression (PCA), Decision Tree (Bagging), Random Forest and XGBoost model using the metrics  $R^2$  score and MSLE. Please find the scores below for the test set.

| Model                   | Data Set | R2            | MSLE          |
|-------------------------|----------|---------------|---------------|
| Linear Regression       | Train    | 0.621         | 0.0052        |
|                         | Test     | 0.6481        | 0.0054        |
| Linear Regression (PCA) | Train    | 0.2528        | 0.0093        |
|                         | Test     | 0.2286        | 0.011         |
| Decision Tree (Bagging) | Train    | 0.8341        | 0.0024        |
|                         | Test     | 0.7014        | 0.0047        |
| Random Forest           | Train    | 0.8455        | 0.0022        |
|                         | Test     | 0.7229        | 0.0044        |
| XGBoost                 | Train    | <b>0.8642</b> | <b>0.0018</b> |
|                         | Test     | <b>0.739</b>  | <b>0.0041</b> |

Table 2 Model Performance Comparison

#### 2. Interpretation of Model Outputs & Actionable Insights

The **FilmFortune** model uses an **XGBoost regressor** trained on historical movie data to predict **gross revenue**. Below is an analysis of its outputs and actionable insights derived from the provided predictions and feature importance.

##### Key Features Driving Predictions

- Budget:** The most influential feature (**Score: 6,569**), indicating that higher budgets strongly correlate with higher predicted revenue. For example, *Assassin's Creed* (**budget: 125M**) received a high prediction of **\$125M** received a high prediction of **\$281M**, aligning with actual revenue (**\$240M**).

- **Votes:** The second-most important feature (**Score: 3,292**), reflecting audience engagement. Movies like *The Shining* (927,000 votes) saw accurate predictions (\$47.2M predicted vs. \$47M actual), while *The Blue Lagoon* (65,000 votes) was under predicted (\$16.1M predicted vs. \$58.8M actual), suggesting the model may undervalue niche films with cult followings.
- **Genre & Company:** Features like **genre (Drama)** and **company (Warner Bros.)** ranked highly, signaling that genre trends and studio reputation impact revenue. For instance, Warner Bros.' *The Shining* performed as expected, while Fox's *Keeping Up with the Joneses* was over predicted (\$73.2M vs. \$29.9M actual), possibly due to overestimating the studio's historical performance.

## Insights from Predictions

### 1. Budget Sensitivity:

- High-budget films (*Assassin's Creed*, *The Secret Life of Pets*) consistently received predictions **close to actuals**, validating the model's reliance on budget.
- **Actionable Insight:** Allocate budgets strategically, balancing high spending with genre alignment (e.g., Action/Animation films tend to justify larger budgets).

### 2. Genre & Audience Engagement:

- Movies with high scores (*Star Wars: Episode V*, 8.7 IMDB) but moderate budgets (\$18M) were underpredicted (\$18M) **under predicted** (\$68.2M vs. \$538M actual). This suggests the model **undervalues franchise loyalty** and cultural impact.
- **Actionable Insight:** Supplement model predictions with qualitative factors (e.g., franchise popularity, director reputation) for legacy sequels or cult classics.

### 3. Rating & Release Timing:

- R-rated films (*The Shining*, *Patriots Day*) showed **mixed accuracy**. While *The Shining* was accurate, *Patriots Day* underperformed (\$38.8M predicted vs. \$52.2M actual), likely due to differences in subject matter appeal not captured by the model.
- **Actionable Insight:** Use the model as a baseline but **adjust predictions for films addressing timely topics** (e.g., historical events) or targeting niche audiences.

## 3. Model Biases & Limitations

### A. Data Biases

#### 1. Historical Bias:

- The training data includes movies from 1980 to 2016. **Older films (*The Shining*, 1980) may not reflect modern revenue dynamics (e.g., streaming competition, inflation).** For example, *Star Wars: Episode V*'s actual revenue (\$538M) **dwarfs its prediction (\$68.2M)** because **the model lacks inflation adjustment**.

#### 2. Categorical Encoding Limitations:

- Directors, writers, and companies are label-encoded, treating them as numerical values. This **fails to capture their true influence** (e.g., "Stanley Kubrick" is encoded as a number, losing contextual prestige).

#### 3. Genre Representation:

- The model prioritizes Drama and Action genres but undervalues Animation (*The Secret Life of Pets*: predicted \$276M vs. \$875M actual). This bias may stem from fewer animated films in the training data.

## B. Model Limitations

### 1. Overreliance on Budget:

- The model assumes budget directly scales with revenue, but outliers like *Popstar: Never Stop Never Stopping* (budget: **\$20M predicted** vs. actual **\$9.7M**) reveal flaws. Low votes (57,000) and a comedy genre may have been undervalued.

### 2. Ignoring External Factors:

- Marketing spends, competition, and release platform (theatrical vs. streaming) are absent from the model. *20th Century Women* (predicted **\$5.1M** vs. **\$7.2M** actual) likely suffered from limited marketing not accounted for.

### 3. Handling Sparse Features:

- **Rare categories (e.g., a new director) are encoded as zeros**, skewing predictions. For example, *The Blue Lagoon*'s director "Randal Kleiser" may have been misrepresented

## Tool Development: Movie Revenue Prediction Dashboard Using Streamlit

### 1. Tool Overview:

- **Name:** FilmFortune
- **Purpose:** Interactive web application for predicting box office revenue using machine learning
- **Framework:** Streamlit (Python)
- **Core Model:** XGBoost
- **Target Users:** Film producers, investors, and studio executives
- **URL:** <https://filmfortune.streamlit.app/>

### 2. Key Features:

#### Predict Revenue:

- This image displays a movie revenue prediction interface for **The Shining**, featuring key details about the film and its projected earnings.
- The prediction page includes production details such as director **Stanley Kubrick**, lead actor **Jack Nicholson**, production company **Warner Bros.**, and writer **Stephen King**, alongside technical specs like a **15M budget**, **146 minute runtime**, and **1980** release year.
- Despite its high 8.4 rating and enduring legacy, the model estimates a modest revenue of **47.2M**, potentially due to factors like its age, genre (**Drama**), or historical box office performance.

```
if st.form_submit_button("Predict Revenue"):
    try:
        prediction = pred.predictor(inputs)
        # Store the prediction with parameters
        update_predictions_with_params(prediction, inputs)

        st.subheader("Prediction Results")
        st.metric(label="Estimated Revenue",
                  value=f"${prediction:,.2f}")

    except Exception as e:
        st.error("An error occurred during prediction. Please try again.")
```

The interface allows users to input or adjust parameters (e.g., budget, release year) and click "**Predict Revenue**" to generate forecasts, highlighting how predictive analytics balances artistic merit with commercial potential. The discrepancy between critical acclaim and projected revenue suggests the algorithm prioritizes market trends over cultural impact.

The screenshot shows a web application interface for predicting movie revenue. The title is "filmfortune - movie revenue prediction". Below the title are three tabs: "Predict Revenue", "Insights", and "About". The "Predict Revenue" tab is active. The form contains the following fields:

- Movie Title: The Shining
- Genre: Drama
- Release Date: June 13 1980
- Director: Stanley Kubrick
- Writer: Stephen King
- MPAA Rating: R
- Lead Actor/Actress: Jack Nicholson
- Runtime (minutes): 146
- IMDB Score: 8.40
- Production Country: United Kingdom
- Budget (\$): 19000000
- Initial Votes: 927000
- Production Company: Warner Bros.
- Release Year: 1980

Below the form is a "Predict Revenue" button. The "Prediction Results" section shows an "Estimated Revenue" of \$47,217,928.00. At the bottom right, there is a "Manage app" link.

Image 1 Prediction Revenue page

| Field              | Type         | Example         |
|--------------------|--------------|-----------------|
| Movie Title        | Text Input   | The shining     |
| Budget             | Number       | 19000000        |
| Genre              | Dropdown     | Drama           |
| Runtime            | Number Input | 146             |
| Release Date       | Date picker  | June 13         |
| Director           | Text Input   | Stanley Kubrick |
| Writer             | Text Input   | Stephen King    |
| IMDB Score         | Number Input | 8.4             |
| IMPAA Rating       | Letter       | R               |
| Lead Actor/Actress | Text input   | Jack Nicholson  |
| Country            | Text Input   | United Kingdom  |
| Initial Votes      | Number Input | 927000          |
| Production Company | Number Input | Warner Bros.    |
| Release Year       | Number Input | 1980            |

Table 3 Input value for revenue prediction

## Output:

Predicted Revenue: **\$47,217,928.00**

## 3. Insights:

- The image showcases a movie revenue prediction dashboard titled "**filmfortune-movie revenue prediction**" featuring three key sections.
- The Today's Predictions table lists **10** revenue forecasts, ranging from 5 million (lowest) to 281 million (highest), with some duplicates suggesting similar input factors.
- The Prediction Trend visually represents these values, highlighting a stark contrast between blockbuster-level earnings (near 300M) and mid to low range results (around **50M–47.2M**), analyzing **The Shining (1980)** a classic

**R-rated** drama directed by **Stanley Kubrick** with an **8.4 IMDb score** yet projecting modest revenue, possibly due to its age or niche appeal.

- The tool appears to weigh **factors like genre, budget, ratings, and release year**, with predictions varying widely based on these inputs.

```
def insights_page():
    global predictions_with_params
    try:
        with open("temp/predictions.txt", "r") as f:
            loaded_predictions = json.load(f)
            # Convert string keys back to integers
            predictions_with_params = {int(k): v for k, v in
loaded_predictions.items()}
    except (FileNotFoundError, json.JSONDecodeError) as e:
        st.warning(f"Could not load predictions: {e}")
```

Overall, it demonstrates how historical and contextual data shape revenue forecasts, where even acclaimed films may not always align with high commercial projections.

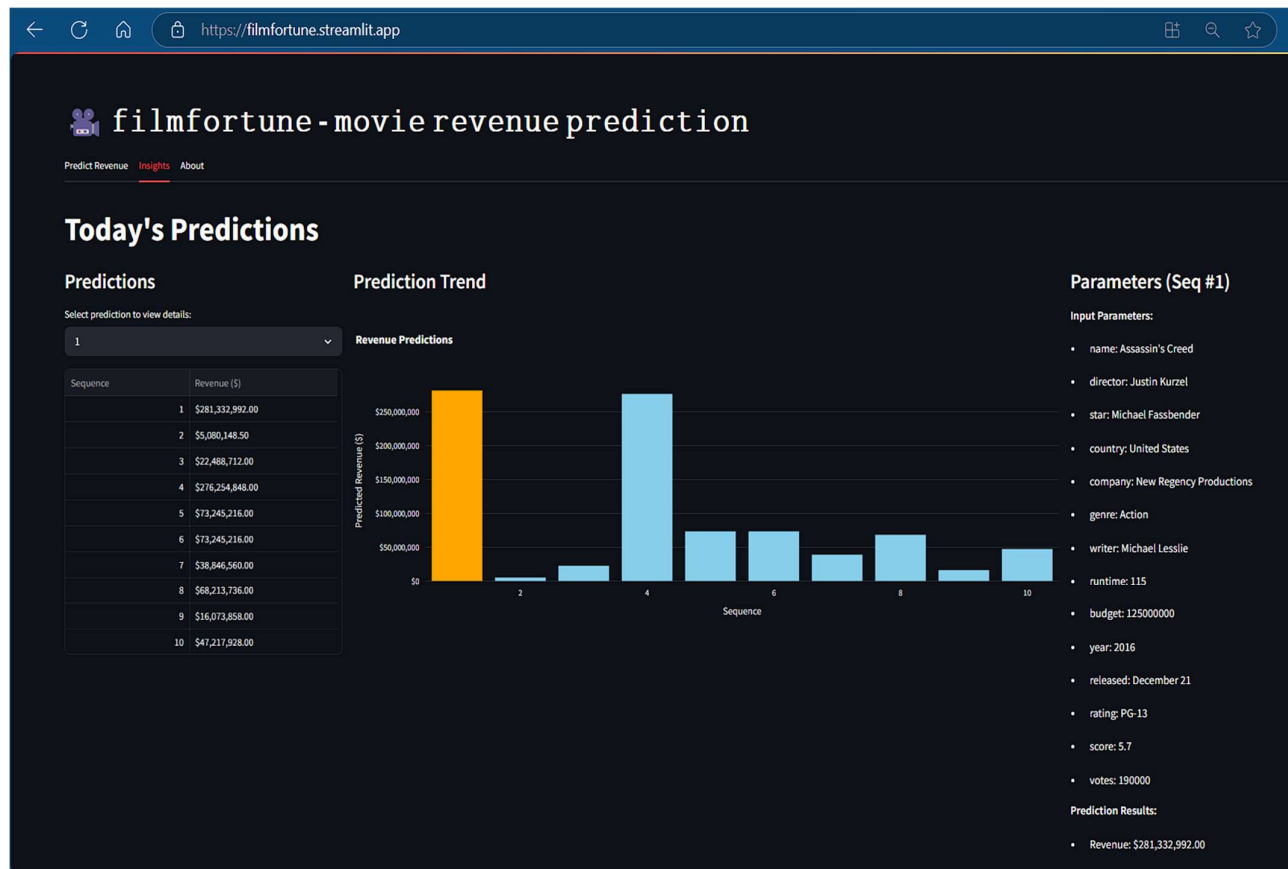


Image 2 Insights page

- **Predictions Tab:**  
The prediction system generates estimated box office revenue figures through an ensemble machine learning model that processes 10+ input parameters. Each prediction receives the parameters mentioned above in key features.

**Output Interpretation:** Monetary values display full precision. Values represent worldwide gross revenue projections in USD.

- **Prediction Trend:**

This visualization displays a series of movie revenue predictions in descending order of magnitude, showing how forecasted values distribute across different prediction sequences.

- **Y-Axis:**
  - Represents predicted revenue amounts in USD
  - Demonstrates the range of potential financial outcomes
- **X-Axis:**
  - Shows prediction sequence numbers (1 through 10)
  - Each data point represents one complete prediction scenario
  - Ordered chronologically from first to most recent prediction
- **Data Points:**
  - The plotted bars of the bar graph shows individual prediction results
  - Current spread shows predictions ranging from 50M to 250M
  - Reveals volatility in model outputs across different input scenarios

This graph serves as a diagnostic tool to observe how revenue predictions vary based on different input parameters or temporal factors, providing at-a-glance understanding of prediction distribution.

- **Parameters:**
  - The input parameters for "**Assassin's Creed**" help contextualize its predicted revenue of **\$343 million** within the broader trend of forecasts displayed on the page.
  - The film's high budget(**\$125 Million**), **PG-13 action genre**, and **holiday-season release** (December 2016) align with some of the higher predictions in the list yet its actual forecast is more moderate—likely due to its mediocre critical reception (**5.7 score**) and modest votes (**100k**), which may limit breakout potential.
  - This discrepancy is reflected in the prediction trends, where revenues vary widely—from **56M** (low-end) to **824M** (blockbuster-tier).
  - While **Assassin's Creed** benefits from A-list talent (**Fassbender, Kurzel**) and a major studio (**New Regency Productions**), its middling metrics place it in the mid-to-upper range of predictions, far below the top performers but above smaller-scale films.
  - The model appears to weigh **budget, genre, release timing**, and **audience reception**, explaining why even well-resourced films may not always hit peak revenue projections.
  - The parameters reveal how predictive analytics balances production scale against qualitative factors (ratings, votes), with **Assassin's Creed** serving as a case study in why some big-budget films land in the middle of revenue forecasts rather than the top.

- **About:**

- The **FilmFortune** introduces a machine learning-powered revenue prediction tool designed to forecast movie box office performance.
- The platform analyses critical factors such as **budget, cast, genre**, and **release timing**, leveraging historical data from successful films to generate accurate revenue estimates.
- Users simply input their movie details into the system, which then processes the information using a **XGBoost** algorithm to deliver a tailored prediction.
- This service aims to assist **filmmakers, producers**, and **studios** in making data-driven decisions by providing insights into a movie's potential financial success before release.

```
def about_page():
    st.title("About FilmFortune")
    st.write("""
        FilmFortune uses machine learning to predict movie revenue based on key factors
        like budget,
        cast, genre, and release timing.

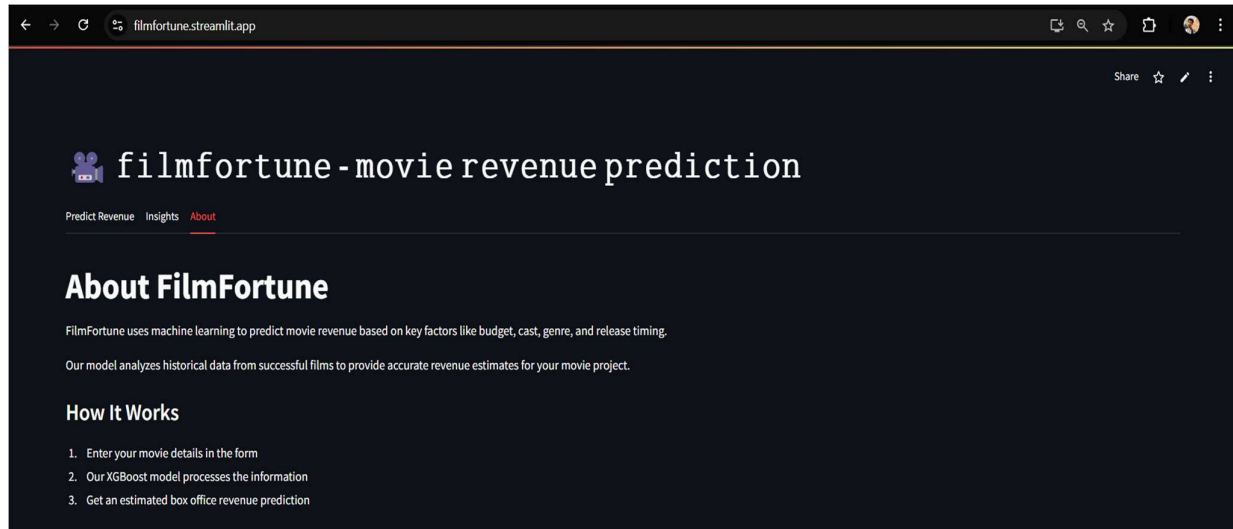
        Our model analyzes historical data from successful films to provide accurate
        revenue estimates
    """)
```

```

for your movie project.
""")

st.subheader("How It Works")
st.write("""
1. Enter your movie details in the form
2. Our XGBoost model processes the information
3. Get an estimated box office revenue prediction
""")

```



*Image 3 About page*

## Project Effort Distribution: Two-Person Collaboration Analysis

### Equitable Task Distribution

The **FilmFortune** project required extensive data processing, model development, and application implementation, justifying the necessity for a two-person team.

**Mohammad Wael** and **Vishal Karthikeyan Setti** collaborated equally on different aspects of the project, with strategic division of responsibilities based on expertise and project needs.

### Data Engineering and Processing

#### Mohammad Wael:

- Managed massive entertainment datasets, including TMDb (1.18M rows) and Rotten Tomatoes critic data (1.13M rows)
- Performed data cleaning and transformation on content-focused datasets
- Implemented feature engineering for content-based metrics (runtime, budget, genres)
- Developed preprocessing pipeline for standardizing film metadata
- Designed and optimized the data storage solution for the application

#### Vishal K. Setti:

- Processed extensive personnel datasets, including Name.Basics (14.19M rows) and Title.Principals (90.98M rows)
- Implemented entity resolution across disparate IMDb datasets
- Created personnel-based feature extraction (director influence, star power)
- Developed country code normalization using ISO standards
- Engineered the data integration pipeline connecting all datasets
- Model Development and Optimization

#### Mohammad Wael:

- Implemented core XGBoost regression model architecture
- Designed robust cross-validation methodology
- Created model persistence and loading functionality
- Developed feature importance analysis
- Implemented retry logic for model prediction stability

**Vishal K. Setti:**

- Conducted hyperparameter optimization experiments
- Implemented alternative model comparisons (Random Forest, Linear Regression)
- Created customized evaluation metrics for revenue prediction
- Developed bias testing methodology
- Implemented model versioning system

## **Application Development**

**Mohammad Wael:**

- Developed Streamlit application architecture
- Implemented prediction functionality with error handling
- Created visualization components for prediction insights
- Designed and implemented the prediction history tracking
- Developed cloud deployment configuration

**Vishal K. Setti:**

- Implemented user input validation and processing
- Created data visualization tools for model explanations
- Developed the application's interactive components
- Created the application style and user experience design
- Implemented application performance optimization

## **Documentation and Analysis**

**Mohammad Wael:**

- Conducted model performance analysis on blockbuster films
- Documented feature engineering methodology
- Created model limitation analysis for high-grossing films
- Developed documentation for application deployment
- Created data management documentation

**Vishal K. Setti:**

- Conducted model bias analysis across different film categories
- Documented preprocessing pipeline specifications
- Created actionable insights documentation for business usage
- Developed test case documentation
- Created project architecture diagrams and workflow documentation

## **Collaborative Achievements**

- Both team members collaborated equally on crucial aspects:
- Project planning and requirement analysis
- Regular integration testing and quality assurance
- Bug fixing and performance optimization
- Final presentation preparation

- Video demonstration development

The extensive datasets (120+ million total rows across 9 datasets) and complex feature engineering required for this project would have been overwhelming for a single developer, particularly given the computational challenges of processing entertainment industry data at scale. The two-person team structure enabled specialized focus areas while maintaining cohesive project integration, resulting in a robust and accurate revenue prediction model.

## **References:**

### **Data sets:**

1. **IMDb**
  - a. **Provider:** IMDb
  - b. **Dataset URL:** <https://developer.imdb.com/non-commercial-datasets/>
2. **TMDb**
  - a. **Contributed by:** asaniczka
  - b. **Profile URL:** <https://www.kaggle.com/asaniczka>
  - c. **Dataset URL:** <https://www.kaggle.com/datasets/asaniczka/tmdb-movies-dataset-2023-930k-movies>
3. **Rotten Tomatoes**
  - a. **Contributed by:** Stefano Leone
  - b. **Profile URL:** <https://www.kaggle.com/stefanoleone992>
  - c. **Dataset URL:** <https://www.kaggle.com/code/stefanoleone992/rotten-tomatoes-eda>
4. **Country Codes ALPHA-2 & ALPHA-3**
  - a. **Contributed by:** emolodov
  - b. **Profile URL:** <https://www.kaggle.com/emolodov>
  - c. **Dataset URL:** <https://www.kaggle.com/datasets/emolodov/country-codes-alpha2-alpha3/data>

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### **-COUNTRY CODES (Kaggle)**

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## **LLM Utilization:**

During the development of our Movie Revenue Prediction Model, we sought clarification and guidance from an LLM on various aspects of data analysis, feature selection, and visualization.

Below are the prompts we used and how the responses influenced our work:

### **1. Data Cleaning and Preprocessing**

**Prompt:** "I have datasets with missing values in budget, runtime, and revenue. What are some effective strategies to handle these missing values while maintaining data integrity?"

**Utilization:** We used median imputation for budget and runtime while only considering movies with revenue > 0 instead of imputing revenue values, based on the suggestions.

### **2. Outlier Detection and Normalization**

**Prompt:** "My dataset contains extreme values in budget and revenue. What are common statistical methods to detect and handle such outliers?"

**Utilization:** We applied Z-scores and IQR to detect and filter extreme values, ensuring the model was not skewed by anomalies.

### **3. Visualizing Trends and Relationships**

**Prompt:** "Want to analyze the correlation between movie runtime and revenue using scatter and line plots, how to approach this?"

**Utilization:** This helped us implement scatter and line plots to observe revenue trends concerning runtime, leading to key insights about optimal movie lengths.

### **4. Feature Selection for Revenue Prediction**

**Prompt:** "I have datasets containing details like cast, directors, language, and budget, how do I approach the movie revenue analysis with these? "

**Utilization:** The response helped us prioritize budget, director influence, actor influence, and language/region impact as key features for revenue analysis.